Advanced Mathematical Techniques Worksheet 9
Dr. Osborne 2021

The purpose of this worksheet is to give you practice with the binomial and multinomial distributions and some applications of the normal distribution to data analysis. You may use a calculator or computer algebra system to do these problems, but please be certain to indicate what you are doing at all times. Please give all answers correct to four digits, so 0.003245786 would be given as 3.246×10^{-3}.

1. In a very strange gambling hall, there is a game in which gamblers can place bets on how many 1's will come up in a run of 24 rolls of a fair six-sided die.
   (a) What is the expectation value of the number of times a 1 will come up?

   (b) What is the probability that 1 will come up its expected number of times?

   Parts (c) – (g) refer to a gambler who places a bet that 1 will come up four times *or more*.
   (c) At the beginning of the run of rolls, what is the probability that the gambler will win?

   (d) Suppose that ten rolls have already taken place and 1 has come up twice. What is the probability that the gambler will win given this information?

   (e) Suppose that ten rolls have already taken place and 1 has come up three times. What is the probability that the gambler will win given this information?

   (f) Suppose that ten rolls have already taken place and none of them resulted in a 1. What is the probability that the gambler will win given this information?

   (g) Explain how the additional information in parts (d) – (f) changes the likelihood of the gambler being successful.

   (h) Suppose now that the die will be rolled 20 times instead of 24. How does this change your answers to (a) and (b)? What does this imply about the statement, "The largest probability is associated with 1 coming up the expected number of times."?

2. You measure a certain length four times and obtain the results 3.26, 3.28, 3.26, and 3.27 cm. Your measurement device indicates a precision of 0.005 cm, but you want to use your repeated measurements to gauge the *actual* uncertainty in this length.
   (a) Use this information to estimate the expectation value and standard deviation of this length.

   (b) Determine the standard deviation of the mean associated with this length and quote your resulting length as "number $\pm$ error".

   (c) Use the normal distribution to estimate the probability that a fifth measurement of this length results in any value ≤ 3.25 cm .

   (d) Suppose a fifth measurement was taken before the results were analyzed and the result was 3.25 cm . Re-do parts (a) – (c) with this new data and comment on any differences seen in the new part (c). How is this situation similar to that found in part (f) on problem 1?

3. A room contains 7 people. You are interested in determining the probability of various distributions of the day of the week on which the people were born. Assume that all days of the week are equally likely.
   (a) Determine the probability that all seven people were born on different days of the week.

   (b) Determine the probability that exactly two people were born on the same day of the week and everyone else was born on different days.

   (c) Determine the probability that exactly two *pairs* of people were born on the same day of the week and everyone else was born on different days.

   (d) Determine the probability that exactly three people were born on the same day of the week and all of the other people were born on different days.

   (e) Show that the sum of the results for parts (a) – (d) is less than 1 and indicate what distributions the remaining probability consists of. Pick one of these and show that its probability is smaller than the difference between your sum and 1.

   (f) Determine the expectation value of the product of the number of people born on Tuesday and the number of people born on Saturday. Why isn't this expectation value just equal to 1? Explain.